Network Layer Security

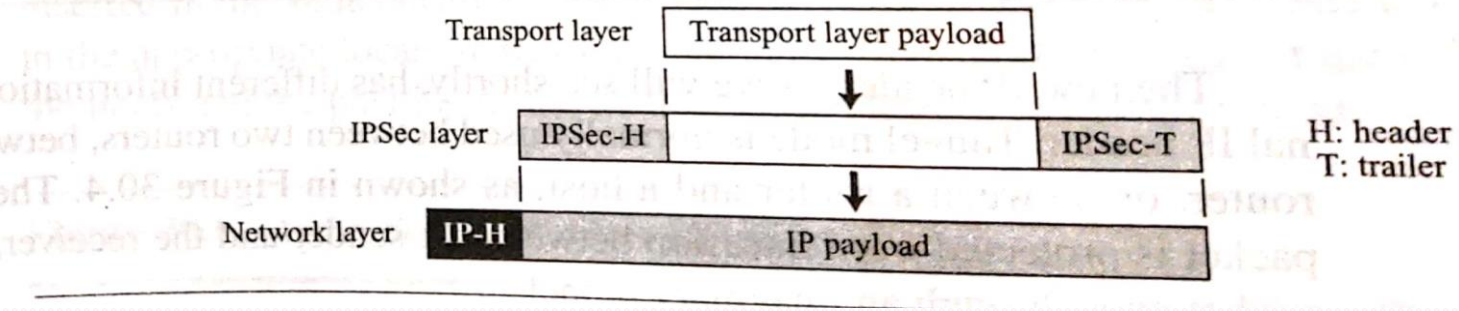
- Need for security at network Layer
  - Not all client/server programs are protected at the application layer
  - Not all client/ server programs at the application layer use the services of TCP to be protected by the transport layer security
  - Many applications like routing protocols directly use the service of IP

IP Security (IPSec)

- Operates in two modes:
  - Transport mode
  - Tunnel mode

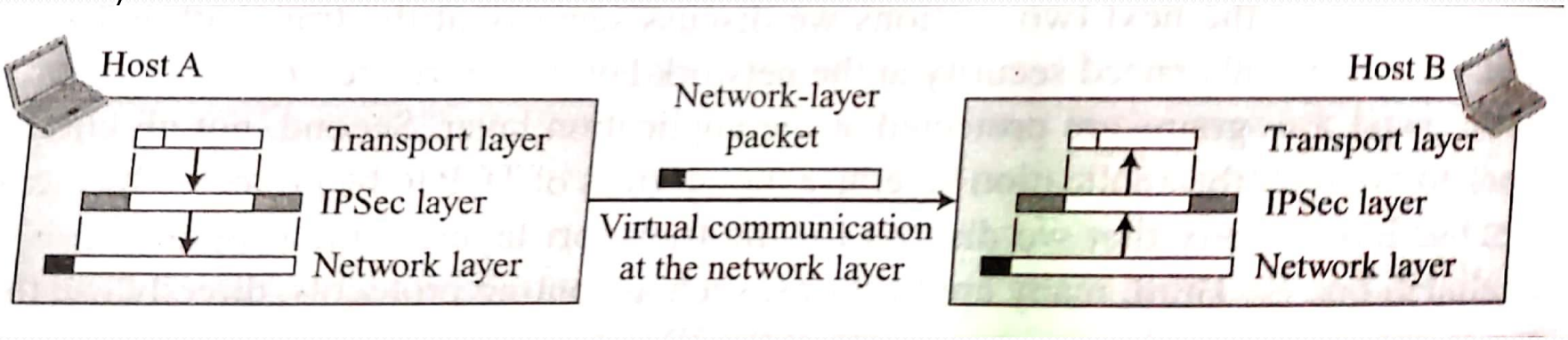
Transport Mode:

- IPSec protects what is delivered from the transport layer to the network layer
- It does not protect IP header
- It does not protect the whole IP packet; it protects only the packet from the transport layer (IP layer payload)
- IPSec header (and Trailer) are added to the information coming from the transport layer



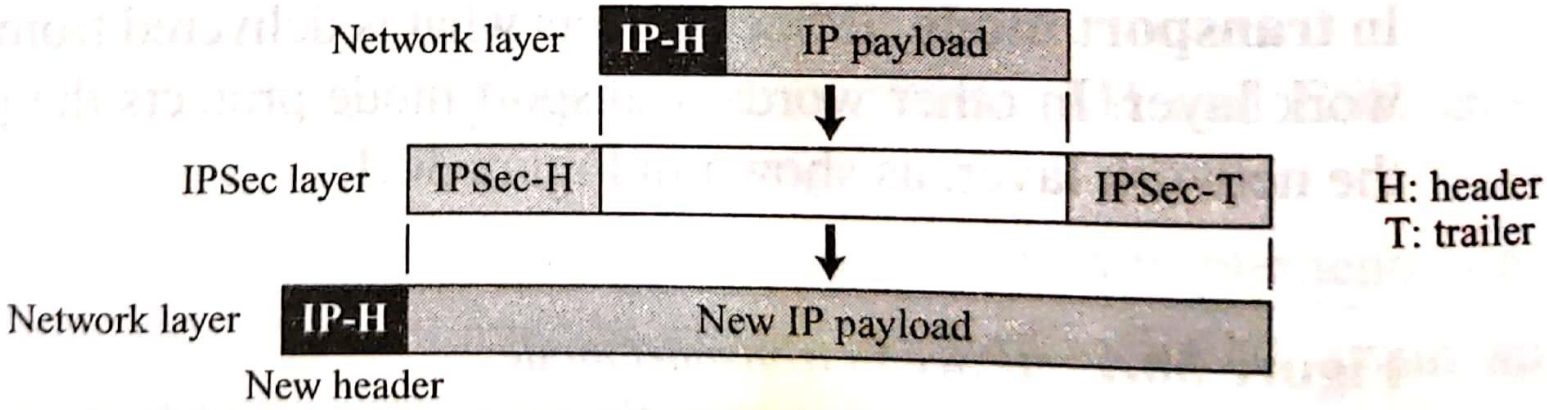
Transport Mode:

- It is normally used when we need host-to-host (end-to-end) protection of data
- The sending host uses IPSec to authenticate and/or encrypt the payload delivered from the transport layer.
- The receiving host uses IPSec to check the authentication and/or decrypt the IP packet and deliver it to the trnasport layer



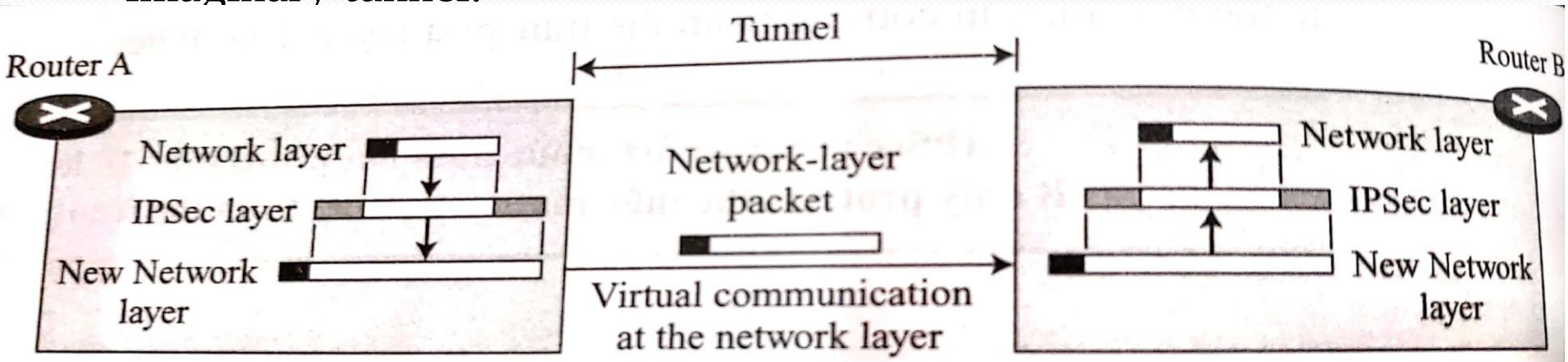
Tunnel Mode:

- It protects the entire IP packet
- It takes IP packet, including the header, applies IPSec security methods to the entire packet, and then adds a new IP header



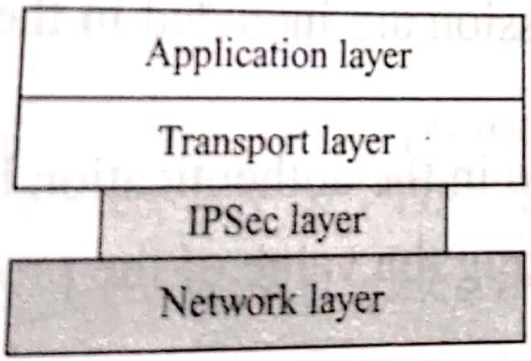
Tunnel Mode:

- New IP header has different information than the original IP header.
- Tunnel mode is used between:
  - Two routers
  - between a host and a router
  - Between a router and a host
- The entire original packet is protected from intrusion between the sender and the receiver, as if the whole packet goes through an imaginary tunnel.

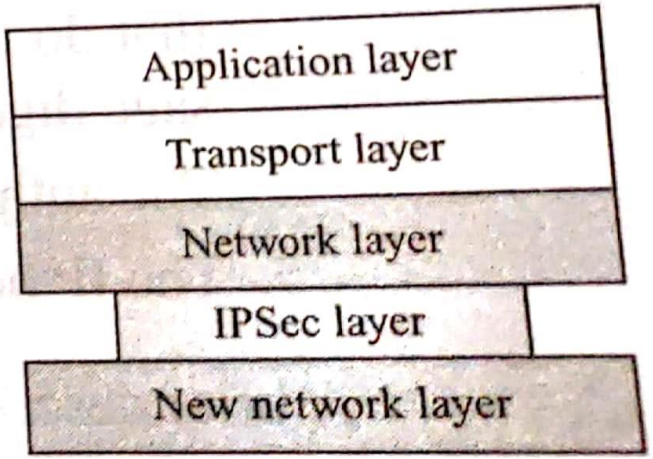


Comparison:

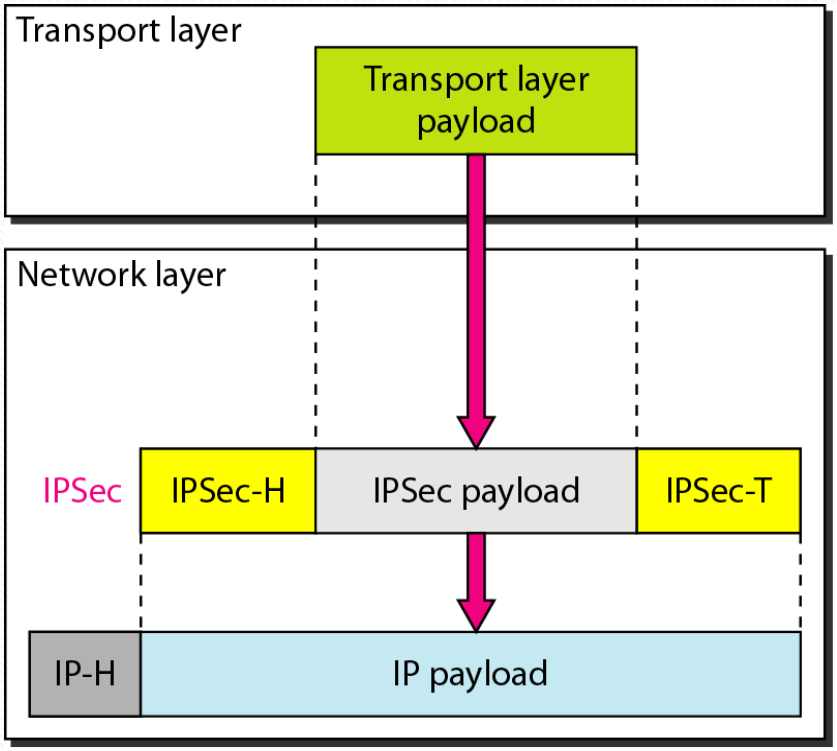
- In Transport mode, the IPSec layer comes between the transport layer and the network layer
- In tunnel mode, the flow is from the network layer to the IPSec layer and then back to the network layer again.



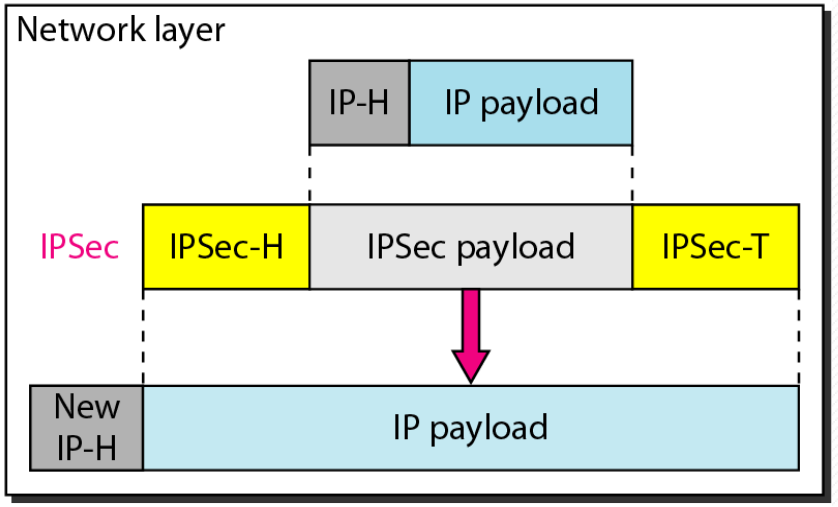
Transport Mode



Tunnel Mode



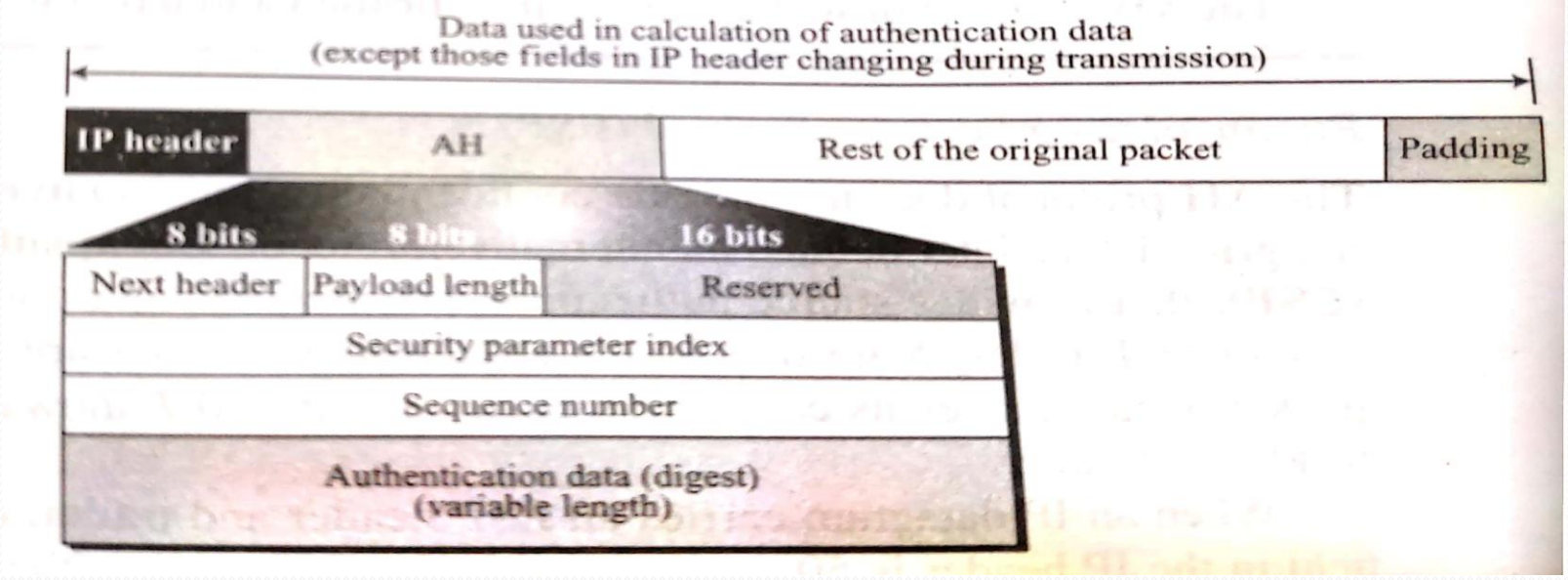
a. Transport mode



b. Tunnel mode

Two Security Protocols

- **Authentication Header (AH):**
  - It is designed to authenticate the source host and to ensure the integrity of the payload carried in the IP packet
  - Uses hash function and a symmetric (secret) key to create a message digest
  - Digest is inserted in authentication header
  - AH is then placed in the appropriate location based on the mode (transport or tunnel)



Two Security Protocols

- When IP sends a AH, the original value in the protocol field of the IP header is replaced by the value 51
- A field inside the authentication header (next header field) holds the original value of the protocol filed (the type of payload being carried by the IP datagram)
- The addition of Ah follows these steps:
  - An AH is added to the payload with authentication data field set to 0
  - Padding may be added to make the total length even for a particular hashing algorithm
  - Hashing is based on total packet
  - Authentication data is inserted in AH
  - IP header is added after changing the value of protocol field to 51.

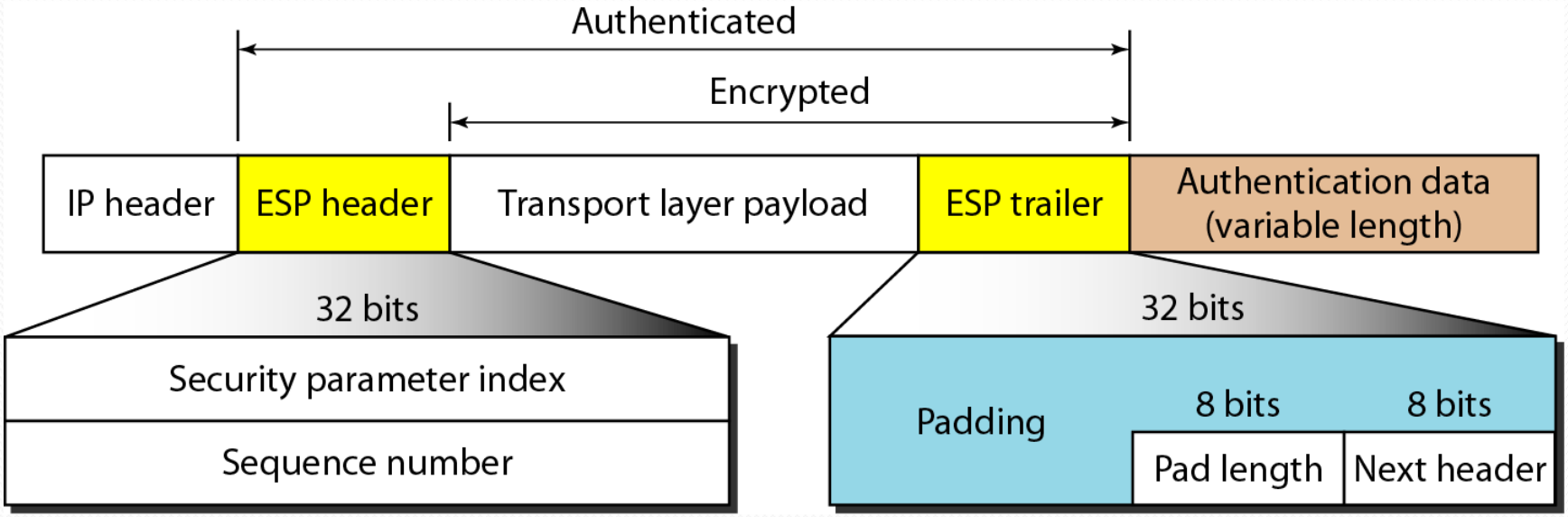
Authentication Header

- **Next header**: the 8-bit next-header field defines the type of payload carried by the IP datagram (such as TCP, UDP, ICMP,..).
- **Payload length**: it defines the length of the authentication header
- **Security Parameter index:** the 32-bit security parameter index (SPI) is same for all packets sent during a connection called a security association.
- **Sequence number:** the 32-bit sequence number provides ordering information for a sequence of datagram
- **Authentication data:** Authentication data field is the result of applying a hash function to the entire IP datagram except for the field that are changed during transit e.g. time-to-live.

Encapsulating Security Payload (ESP):

- Since AH does not provide privacy, IPSec later define an alternative protocol that provides source authentication, integrity, and privacy called **Encapsulating Security Payload (ESP)**
- ESP adds a header and trailer**.**
- ESP authentication data is added at the end of the packet, which makes calculation easier
- When an IP datagram carries an ESP header and trailer, the value of the protocol field in the IP header is 50

Encapsulating Security Payload (ESP):



Header and Trailer are as follows:

- **Security parameter index:** the 32-bit security parameter index field is similar to that defined for the AH protocol.
- **Sequence number:** the 32-bit sequence number field is similar to that defined for the AH protocol.
- **Padding:** this variable-length field (0 to 255 bytes) of 0s serves as padding.
- **Pad length:** the 8-bit pad length field defines the number of padding bytes
- **Next header:** the 8-bit next-header field is similar to that defined in the AH Protocol.
- **Authentication data:** it is the result of applying an authentication scheme to part of the datagram.

AH Versus ESP

- The ESP Protocol was designed after AH Protocol was already in use.
- ESP does whatever AH does with additional functionality (privacy).
- Why do we need AH ?
  - We don’t, but the implementation of AH is already included in some commercial products.

Services Provided by IPSec

The two protocols AH and ESP can provide several security services for packets at the network layer as shown in the table below:

Services	*AH*	*ESP*
Access control	Yes	Yes
Message authentication (message integrity)	Yes	Yes
Entity authentication (data source authentication)	Yes	Yes
Confidentiality	No	Yes
Replay attack protection	Yes	Yes

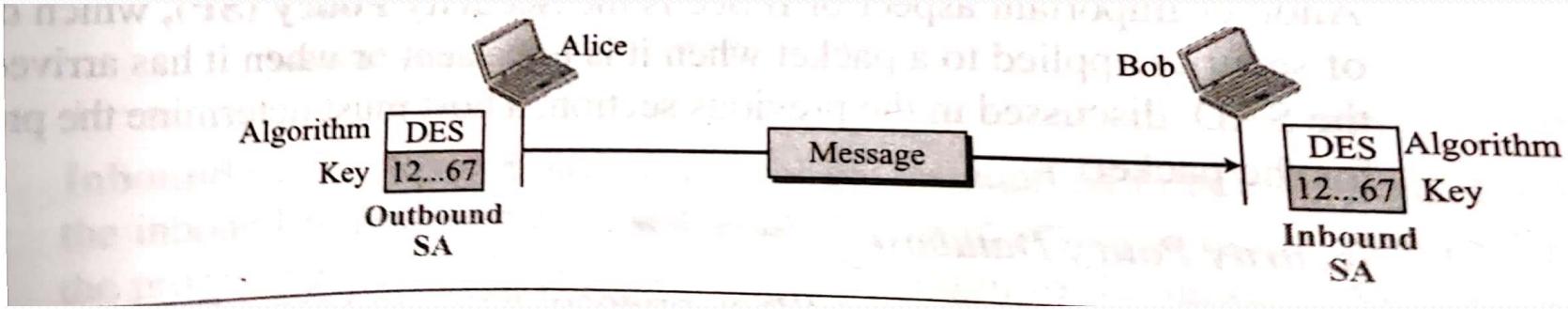
Services Provided by IPSec

- **Access Control:** IPSec provides access control indirectly by using a Security Association Database (SADB).
- **Message Authentication:** the integrity of the message is preserved in both AH and ESP by using the authentication data.
- **Entity Authentication:** The security association and the keyed-hashed digest of the data sent by the sender authenticate the sender in both AH and ESP.
- **Confidentiality:** The encryption of the message in ESP provides confidentiality. AH doesn't provide confidentiality.
- **Replay Attack Protection:** both protocols prevent replay attack by using sequence numbers.

Services Association

- It is a mechanism that IPSec used to establish the security parameters.

- IP is connectionless protocol (each datagram is independent of others).



Services Association Database (SAD)

- What if Alice needs to send to many people and receive from many people too.
- She needs to have multiple inbound and outbound SAs.
- Thus, SADB is needed to collect those se of SAs.
- SADB it is a two-dimensional table with each row defining a single SA.
- Normally, there are two SADBs one inbound and one outbound.

Services Association Database (SAD)

Index	SN	OF	ARW	AH/ESP	LT	Mode	MTU
< SPI, DA, P >							
< SPI, DA, P >							
< SPI, DA, P >							
< SPI, DA, P >							

Security Association Database

Legend:

SPI: Security Parameter Index
DA: Destination Address
AH/ESP: Information for either one
P: Protocol
Mode: IPSec Mode Flag
SN: Sequence Number
OF: Overflow Flag
ARW: Anti-Replay Window
LT: Lifetime
MTU: Path MTU

Security Policy

- Defines the type of security applied to a packet when it is to be sent or when it has arrived
- Before using SAD a host must determine the predefined policy for the packet.

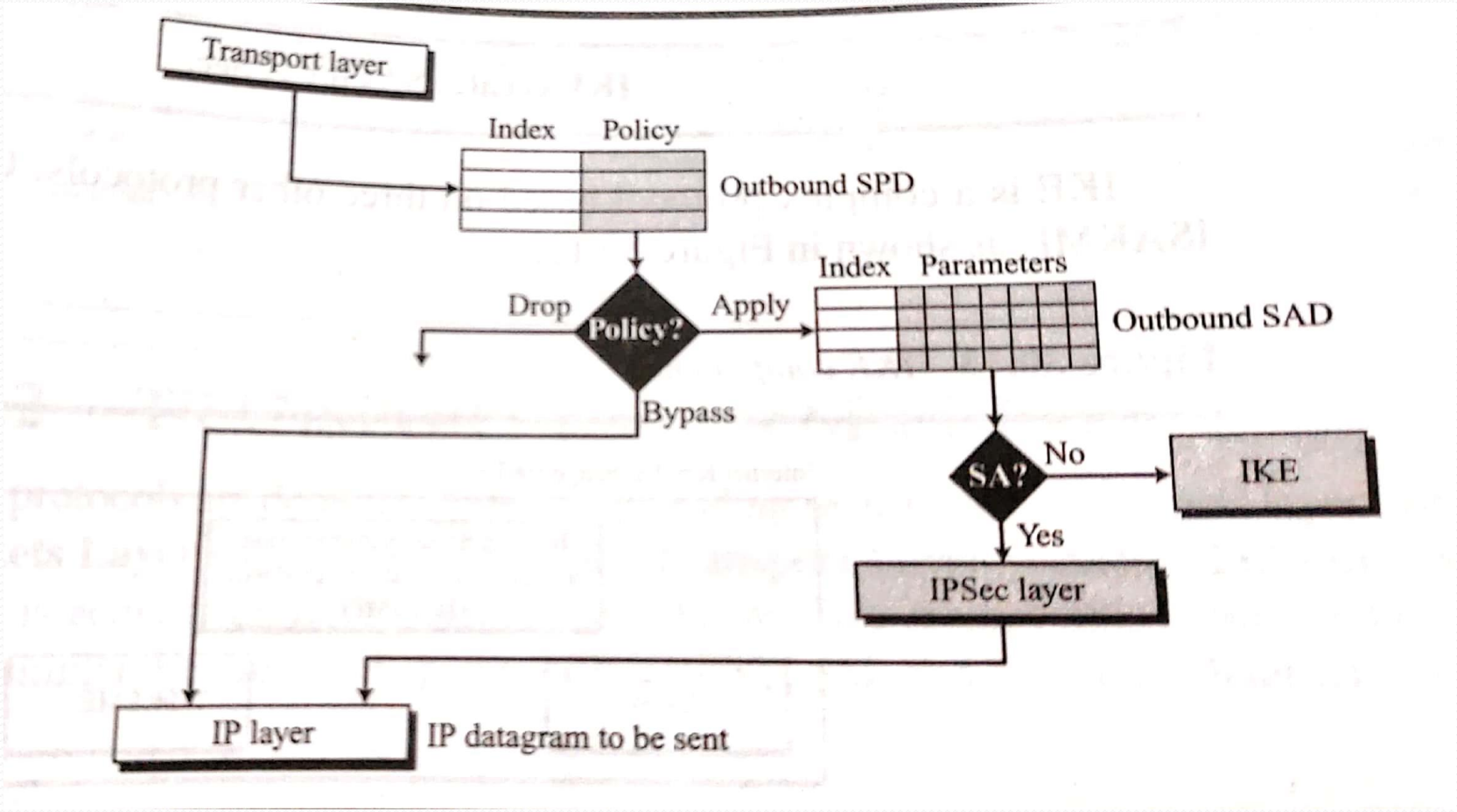
Security Policy Database (SPD)

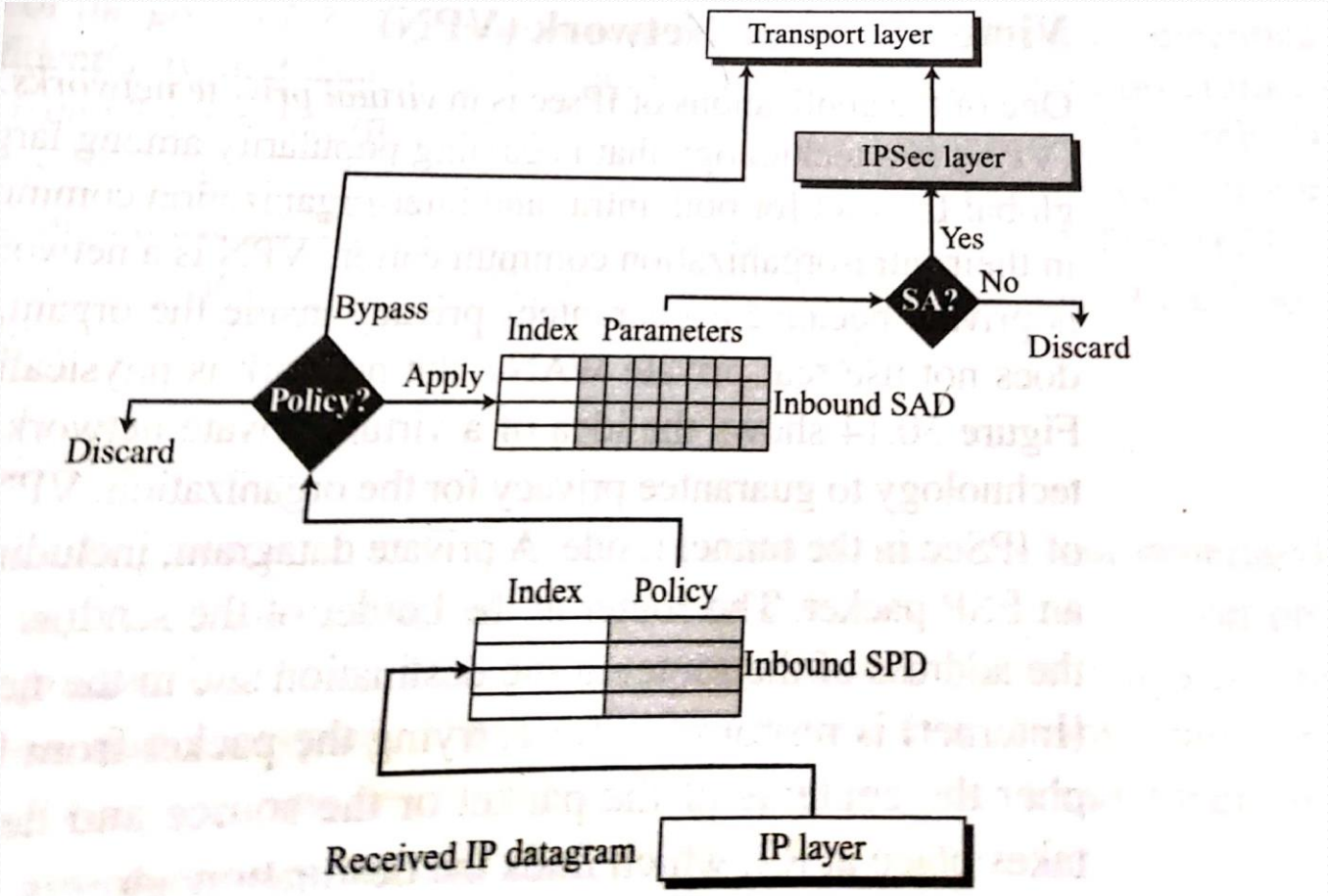
- Each host that is using IPSec protocol needs to keep a Security Policy Database (SPD).
- Need for inbound SPD and outbound SPD

Index	Policy
< SA, DA, Name, P, SPort, DPort >	
< SA, DA, Name, P, SPort, DPort >	
< SA, DA, Name, P, SPort, DPort >	
< SA, DA, Name, P, SPort, DPort >	

Legend:

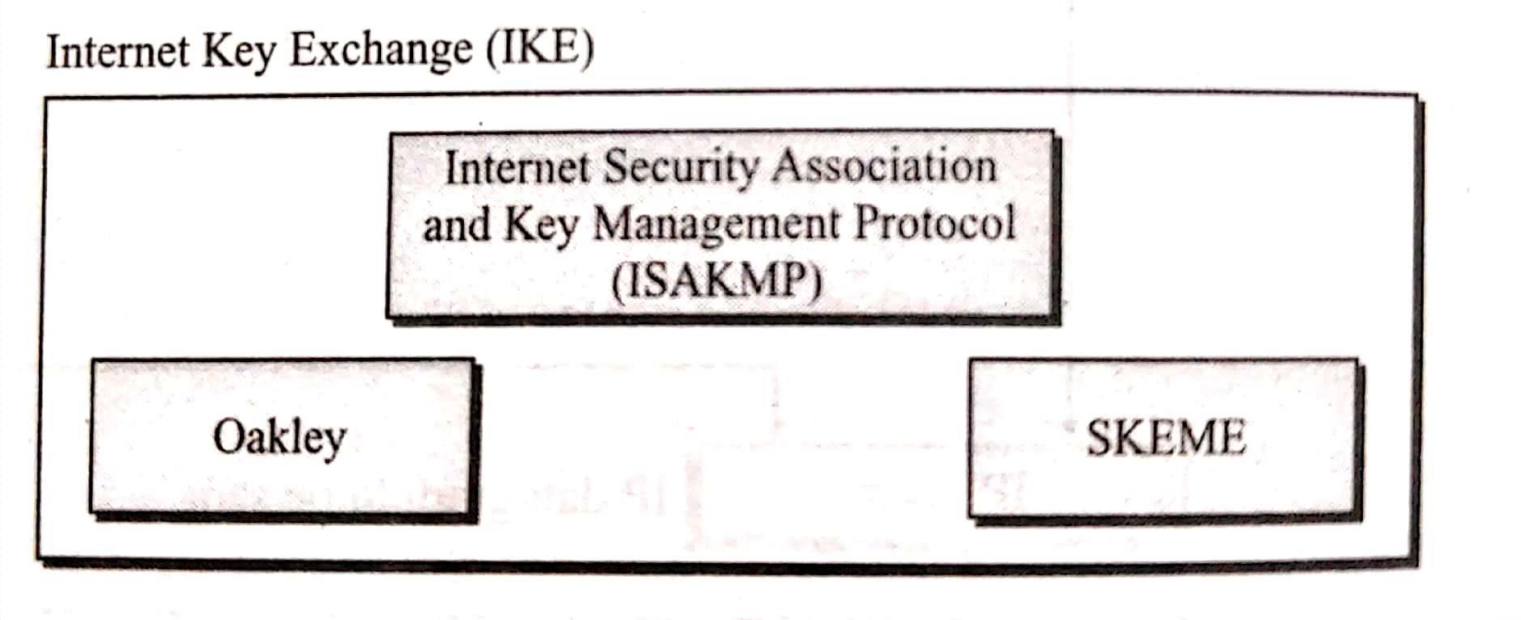
SA: Source Address
DA: Destination Address
P: Protocol
SPort: Source Port
DPort: Destination Port


Transport layer
Index
Policy
Outbound SPD
Index
Parameters
Drop
Policy?
Apply
Outbound SAD
Bypass
SA?
No
IKE
Yes
IPSec layer
IP layer
IP datagram to be sent


Transport layer
IPSec layer
Yes
SA?
No
Discard
Bypass
Index Parameters
Policy?
Apply
Inbound SAD
Discard
Index Policy
Inbound SPD
Received IP datagram
IP layer

Internet Key Exchange (IKE)

- Oakley is a key creation protocol
- SKEME is also a key generation protocol
- Internet Security Association and Key Management Protocol (ISAKMP) actually implements the exchanges defined in IKE



Virtual Private Networks

